

5. Priority Scheduling (Non-Preemptive)

```
#include <stdio.h>
int main()
{
    int n, bt[20], pr[20], wt[20], tat[20], done[20]={0}; float
    awt=0, atat=0;
    printf("Enter number of processes: "); scanf("%d", &n);
    for (int i=0;i<n;i++){ printf("BT and Priority of P%d: ", i+1);
    scanf("%d %d", &bt[i], &pr[i]); }
    int time=0;
    for (int c=0;c<n;c++){
        int idx=-1;
        for (int i=0;i<n;i++) if(!done[i] && (idx==-1 ||
pr[i]<pr[idx])) idx=i;
        wt[idx]=time; time+=bt[idx]; tat[idx]=time; done[idx]=1;
        awt+=wt[idx]; atat+=tat[idx];
    }
    printf("P\tBT\tPR\tWT\tTAT\n");
    for (int i=0;i<n;i++) printf("P%d\t%d\t%d\t%d\t%d\n", i+1,
bt[i], pr[i], wt[i], tat[i]);
    printf("Average WT = %.2f, Average TAT = %.2f\n", awt/n,
atat/n);
    return 0;
}
```

Sample Input: 3 / 5 2 / 3 1 / 8 3

Output:

```
Enter number of processes: BT and Priority of P1: BT and Priority of
P2: BT and Priority of P3: P      BT      PR      WT      TAT
P1   5       2       3       8
P2   3       1       0       3
P3   8       3       8      16
Average WT = 3.67, Average TAT = 9.00
```